

MPLAD-TRACE 360™ TECHNICAL ARCHITECTURE & EXECUTIVE DOSSIER

Field Inspections, AI Verification, Material Quality Standards & Fiscal Circuit-Breaker Guide

Operational Reference for Evaluating Public Works, Quality Standards & Fund Flow Controls

1. System Architecture & Lifecycle of an Inspection

When an inspecting officer captures a field inspection via the Mobile PWA, a dual-layer statutory workflow executes automatically to eliminate ghost billing, unauthorized geographical uploads, and contractor collusion.

Phase	Component	Function & Technical Enforcement
1. Capture	Hardware GPS & Camera	Enforces active device location services before opening camera viewfinder. Cryptographically stamps coordinates directly into image canvas.
2. Walkthrough	360° Video Recording	Mandatory continuous panoramic video walkthrough using MediaRecorder API to prove active work perimeter and true site physical progress.
3. Offline Cache	Encrypted Outbox	Stored locally in browser IndexedDB/localStorage with offline cryptographic hash if deep rural 4G/Wi-Fi is disconnected.
4. Cloud Sync	FastAPI Ingestion	Transmits package via REST API to central backend once connectivity resumes; logs immutable SHA-256 Audit Trail.
5. AI Verification	Multi-Stage Vision & Risk	Haversine distance geofencing (<=150m tolerance), OpenCV structural feature matching, and BoQ supply reconciliation.
6. Fiscal Lock	PFMS Circuit-Breaker	Compares physical progress observed against financial burn. Triggers escrow freeze if discrepancy exceeds 15% or status is Stalled.

2. How Does AI Evaluate Construction Material Quality?

A common technical inquiry is: *'Since AI software cannot physically touch or squeeze concrete, steel, or sand, how can it reliably verify material quality?'*

In MPLAD-TRACE 360, material verification is executed through **Three-Way AI Triangulation** rather than subjective visual guessing:

A. Computer Vision Surface & Texture Analytics

- **Concrete Segregation & Honeycombing:** Improper water-cement ratios or inadequate vibrator compacting leave hollow pockets and exposed stone voids. OpenCV Laplacian edge filters detect surface porosity; void depths exceeding 5mm are flagged as *Substandard Concrete Vibration*.
- **TMT Steel Rebar Oxidation:** Color histogram segmentation analyzes exposed rebar cages for rust scale. Oxidation exceeding 15% on tensile bars flags *Structural Corrosion Alert*.
- **Efflorescence & Salinity:** White salt blooming on new brickwork indicates high-saline mixing water or low-grade kiln bricks. Color-clustering filters flag masonry durability failure.

B. Digital Concrete Cube Test Verification (IS 516 / IS 456 Standards)

Indian Standards mandate casting 150mm x 150mm test cubes during casting. These are cured and crushed in a Compressive Testing Machine (CTM) at 7 and 28 days. The inspecting officer logs the verified compressive strength (e.g. 24.8 MPa for M20 concrete). The AI evaluates this against standard curing curves; if strength falls below 85% of specified design strength, the material is rejected and the stage payment is halted.

C. GeM & GST e-Way Bill Volumetric Cross-Reconciliation

Contractors may claim Grade 53 OPC cement on paper but use adulterated fly-ash on-site. The AI cross-checks the Bill of Quantities (BoQ) volume against verified delivery records on Government e-Marketplace (GeM) and GST e-Way bills. If casting a 3,000 sq.ft hall requires 450 cement bags but supplier records only reflect 180 bags delivered to that coordinate, the system detects a **60% Material Volumetric Deficit**, exposing material dilution without laboratory delay.

3. When Does the AI Freeze Funds? (Risk Thresholds)

The automated circuit-breaker monitors the 7-factor MoSPI Risk Matrix (0-100 score). Depending on composite risk and specific red-line anomalies, fund flows are regulated as follows:

Score Range	Risk Level	Disbursal Status	System Action & Fiscal Mechanism
0 – 29	NORMAL	Active (100%)	Tranche releases pass smoothly via PFMS escrow upon milestone sign-off.
30 – 49	WATCH	Permitted	Payments released; 14-day expedited field verification scheduled.
50 – 74	HIGH RISK	Conditional Hold	Escrow disbursal queued; requires District Planning Officer (DPO) re-audit.
75 – 100	CRITICAL	FROZEN (HALTED)	Zero Ghost Billing Circuit-Breaker triggers lock. Payment button disabled. Tranche release withheld.

Immediate Automatic Freezing Triggers (Independent of Score):

- Progress Discrepancy > 15%:** When claimed financial burn outpaces verified physical execution (e.g. 75% payment claimed vs 30% substructure built).
- Stalled Work Status:** Any inspection logging prolonged abandonment or contractor demobilization.
- Substandard Material Failure:** Failed 28-day concrete cube test or counterfeit reinforcement.
- Active Contract Dispute:** Unresolved contractor claim or arbitration flagged in ledger.

4. Where to Check & How to Unfreeze Halted Funds

Where to Verify in the Web Application:

- Tab 3: PFMS Fund Flow & Ledgers (Project Detail Page):** Displays the interactive 6-stage pipeline. If frozen, a prominent red alert appears: *'PFMS ESCROW DISBURSAL HALTED: DISCREPANCY FLAGGED'* and Step 5 (Paid) switches to **Halted**. The payment release button is locked with **Tranche Payment Locked (Awaiting DC Clearance)**.
- AI Risk & Anomaly Center:** Lists all GHOST_BILLING_SURVEILLANCE alerts with exact variance (e.g. Δ 42% discrepancy) and contractor risk profile.
- Tab 4: Field Evidence & Photos:** Shows geotagged before/after photos, AI CV consistency %, and the 360° Site Video Walkthrough player.

Step-by-Step Protocol to Unfreeze Funds:

Step	Responsible Party	Action Required
Step 1: Rectification	Executing Contractor	Execute physical work to catch up with financial milestone (e.g. cast columns/lintels) or recast defective concrete.
Step 2: Re-Inspection	Field Inspecting Officer	Conduct fresh site inspection with hardware GPS and record mandatory 360° video. Log progress = 75% (Satisfactory).
Step 3: Alert Resolution	District Planning Officer (DPO)	Access 'AI Risk & Anomaly Center', click 'Take Action' -> 'Resolve Alert', and enter official re-verification notes.
Step 4: AI Re-Assessment	Risk Engine (Automated)	System re-evaluates risk: Discrepancy drops to 0%, composite score drops below 30 (NORMAL). Red lock clears automatically.
Step 5: Release Resumed	PFMS Gateway (Automated)	Escrow disbursal unlocked. Tranche release button becomes active for authorized finance officers.

Key Takeaway for Project Teams: The MPLAD-TRACE 360 platform eliminates ghost billing and substandard construction by replacing unverified self-certification with objective 3-way data triangulation (Geotagged 360° Video + Lab Cube Tests + GST/GeM supply records). Funds are never locked arbitrarily—they freeze automatically at Risk Score $\geq$ 75 or Milestone Delta > 15%, and unfreeze systematically upon verified physical re-inspection.